TripPlanner software docs 1

TripPlanner System Documentation

github link: https://github.com/ABC2023/TripPlanner
generated link: https://github.com/ABC2023/TripPlanner/blob/main/docs/UC10_TripPlanner.pdf

NOTE: for alternative images look for the deliverables folder in the repository

1. Introduction

1.1 Purpose of the Document

This document provides a comprehensive technical specification of the TripPlanner system, including its architecture, design patterns, and core features.

Author: Cristhian Leal-Mora Rodríguez

1.2 System Overview

TripPlanner is a collaborative trip planning application designed to help groups organize shared travel itineraries.

Core Functionalities:

- Trip creation and collaborative project management
- Room creation and collaboration by planning
- Point of interest (POI) database with filtering capabilities
- Interactive map visualization with Leaflet.js
- Crowd-based trip discovery with ranking and voting capabilities
- Dual interface support (CLI and JavaFX GUI)
- Persistence via JSON serialization (GSON) and SQLITE DATABASE, IN-MEMORY)

Application Type: Desktop application with CLI and GUI interfaces
Target Users: Groups of travelers, trip organizers, and participants

1.3 Technical and Software Requirements

Software Requirements

Component	Specification
Programming Language:	Java 23
Build System:	Gradle Wrapper
UI Framework:	JavaFX 21
Maps:	Leaflet.js 1.9.4
Serialization:	Gson 2.10.1
Database Driver:	Microsoft JDBC Driver 3.3.0
Database Engine:	Microsoft SQL Server
Operating System:	Windows

Hardware Requirements

Platform-independent (standard Java application requirements apply)

1.4 Related or Comparable Systems

Similar collaborative trip planning applications exist in various forms.

2. User Stories

Terms with additional definitions at the bottom are marked with asterisks (*)

US1: Room Creation

As a room creator
I want to create a room
So that I can organize a collaborative trip

US2: Search Points of Interest

As a room member
I want to search by name, rating, tags, and location
So that I can find relevant places to add to the trip

US3: Propose Event Addition

As a room member
I want to propose a POI as an event
So that the group can consider including it in the trip

US4: Collaborative Proposals

As a room member
I want to see proposals made by other members
So that accurate suggestions are prioritized

US5: Create Alternative Branch

As a room administrator
I want to create alternative branches* in the trip graph
So that we can explore different plan options

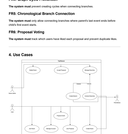
US6: Connect Branches

As a room administrator
I want to connect branches* in the trip graph
So that we can build complete multi-path itineraries

3. Dictionary

Term	**Definition**
Room	Logical group with one administrator and many members, each has a personal trip plan
Point	Point of interest - object representing points during trip (restaurants, monuments, roads, etc.)
Rating	Numeric value from 1 to 5 stars
Tags	Keywords: FOOD, CULTURE, OUTDOOR, NIGHTLIFE, FOCUS - SIGHTSEEING, SCIENCE***
Event	Point in a specified sequence of events in chronological order
Branch	Trip for room members to propose trip changes, compared by administrators
Trip Graph	Trip state where members plan to be at a POI between defined start/end times

4. Use Cases



5. Internal Steps: UC10 - Propose Add Event

Primary Actor

Room Member

Preconditions

- User is logged in
- User has selected a room

Main Flow

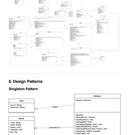
1. User selects "propose new event" option
2. User selects target branch from list
3. System displays POI search interface
4. User optionally enters filter criteria
5. System filters POIs based on criteria and room destination
6. System displays results
7. User selects POI from filtered list
8. System displays POI details
9. User enters start time
10. User enters end time
11. System validates time inputs
12. System creates proposal with selected branch, POI, and interval
13. System adds proposal to branch
14. System displays success message

Exceptional Flows

Condition	System Response	Next Action
User cancels search	Return to branch selection	Return to step 2
No criteria entered	Show all POIs	Continue to step 6
No results found	Allow proposal creation, schedule after last event	Return to step 10
Invalid time format	Throw IllegalArgumentException	Return to step 10
End time before start time	Throw IllegalArgumentException with error message	Return to step 10
Branch deleted while creating session	Return Main Menu	Log, setBranch(null)

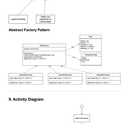
6. Storyboards

Room Management Interface



Trip Management Interface

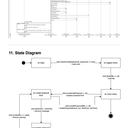
Create Event Proposal Interface



7. Class Diagram

8. Design Patterns

Singleton Pattern



Decorator Pattern

State Pattern

Abstract Factory Pattern

9. Activity Diagram

10. Sequence Diagram

11. State Diagram